# Data Types

Comprehensive breakdown: Structure, Representation, Nature, and Labeling for Data Science.

# What is Data?

## Raw Information & Facts

Data is the raw collection of facts, figures, symbols, or observations representing events or phenomena.

## Transformation to Insights

When organized, processed, and analyzed, data transforms into valuable insights that drive decision-making and ML models.

# 3 Ways to Categorize



## 1. By Structure

Structured, Unstructured, and Semi-Structured formats.



## 2. By Nature

Quantitative (Numeric) vs Qualitative (Categorical) values.



## 3. By Labeling

Labeled vs Unlabeled data for Supervised / Unsupervised ML.

# Data Types Based on Structure

## 1. Structured Data

Organized in strict tabular rows & columns (Databases, Spreadsheets).

**Examples:** Sales records, customer details, bank transactions

## 2. Semi-Structured Data

Contains tags or organizational markers without rigid tabular schemas.

**Examples:** JSON, XML files, NoSQL DBs, sensor logs

## 3. Unstructured Data (80% of global data)

Raw information with no predefined structure; hard to process directly.

**Examples:** Images, videos, text files, audio recordings

CLASSIFICATION 2

# Data Types Based on Nature

## Quantitative Data

Numeric, objective, and measurable values analyzed statistically.

- **Discrete:** Whole counts (e.g., number of store customers).
- **Continuous:** Measured values (e.g., age, temperature, revenue).

**Usage: Finance, Research & Engineering**

## Qualitative Data

Descriptive, non-numeric information describing qualities or categories.

- Focuses on attributes, opinions, and characteristics.
- **Examples:** Customer feedback ("satisfied"), colors, transcripts.

**Usage: Behavior Analysis & Surveys**

# Numerical, Categorical & Ordinal Data

## Numerical Data

Features measured strictly in numbers.

**Examples:** Age, Income, Height, Salary.

## Categorical Data

Represents discrete labels without order.

**Examples:** Gender, Fruit type, Country.

## Ordinal Data

Categorical data with an essential ranking order.

**Examples:** Clothing sizes (S, M, L), Education level.

# Data Types Based on Labeling

## 1. Labeled Data

Contains both input variables (features) AND target output answers.

**Example:** Features ("Age", "Income") linked to Target ("Loan Approval Status").

**Used in: Supervised Learning**

## 2. Unlabeled Data

Contains only input variables without any target outputs or answers.

**Example:** Raw customer logs or unannotated images.

**Used in: Unsupervised Learning & Clustering**

# Summary: Data Types & Machine Learning Tasks

| Category | Data Subtypes | Common Examples | ML Algorithms / Tasks |
|---|---|---|---|
| **Structure** | Structured / Unstructured / Semi | SQL tables, Images, JSON | XGBoost, CNNs, NLP models |
| **Nature** | Quantitative vs Qualitative | Revenue, Feedback, Colors | Regression, Sentiment Analysis |
| **Representation** | Numerical / Nominal / Ordinal | Age, Fruit type, Clothing size | One-Hot / Ordinal Encoding |
| **Labeling** | Labeled vs Unlabeled | Features+Target vs Raw Data | Supervised vs Unsupervised |

# Thank You!